

Question # 01**[20+10+10=40 Points]**

A. Find out logical & syntax error. For each error, mention either logical or syntax & correct the error without any reasoning: [10+10=20 Points]

```
void print1(int x[], const int SIZE){
    ...
}
void print2(int x[][5], const int NROWS){
    ...
}
void print3(int **x, const int SIZE){
    ...
}
int main(){
    int a[10]={...}; //Array has 10 values
    int b[3][5]={...}; //Array has 15 values
    print1 (b, 15);

    print1 (a, 15);

    print1 (b[0], 15);

    print2 (a, 2);

    print3 (b, 15);

    print3 ((int**)b , 15);
    -----
int g=5;

class A{
    int x;

public:
    int y=5;

    A& A(){    x=g;  }

    A(const A a){    ...  }

    A operator *= (const A&);

    A& operator / (const A& a){
        A newA;
        ...
        return newA;
    }
};
```

B. Suppose your program contains the following class definition and suppose the main function of your program contains the following declaration: **[10 Points]**

```
Car hyundai(500...), jaguar(700...), cultus(600...);
```

Which of the following statements are allowed in the main function of your program? (For **legal** statements write 'Y' and 'N' for **illegal**?)

<pre>class Car{ double price, profit; double getProfit(); public: Car (int); Car (const Car &); void setPrice(double); void setProfit(double); double getPrice(); double operator-(Car &); Car operator+(Car &); };</pre>	
Statements	Legal [Y/N]
jaguar.setPrice(30.97);	
jaguar.getPrice();	
jaguar.getProfit();	
cout<<"Price:"<<price<<'\n';	
hyundai.setPrice(jaguar.getPrice());	
hyundai = jaguar;	
Car *p=new Car;	
float diff = jaguar - hyundai;	
hyundai.price = 4999.99;	
Car newC=cultus+ hyundai -jaguar;	

C. For each statement in main point an arrow towards relevant member Function: **[10 Points]**

<pre>class Car{ double price, profit; public: static int factor; Car (); Car (int); Car (int, int); Car (const Car &); Car operator-(Car &); Car operator--(Car &); Car operator+(Car &); Car operator+=(Car &); Car operator+(double); Car& operator=(Car &); Car& operator++(); Car operator++(int); ~Car(); };</pre>	<pre>int main(){ Car c1; Car c2(3,4); c2=c1; c1++; Car c3=c1; Car *c=new Car(5); c1+57345.485; c3-c2; delete c; cout<<c1::factor<<'\n'; new Car;</pre>
---	--

Write code/functions for the following descriptions.

- A. Write global function "**add**" to add 2 objects of Point class and return a new object of Point class. [5]

```
class Point {  
    int x, y;  
public:  
    //constructors & getter  
    //setter given here  
};
```

- B. Provide the implementation of the following class named **Fuzzy**. Each **data member** of this class should contain **2, 1 or 0 (default value is 0)** for a particular object. [3+3+4+2+3=15 Points]

```
class Fuzzy{  
    int x;  
    int y;
```

//Write one set function to set value of both data members

//Write a non-parameterized constructor to set default value of DM's, call set function inside

//overload minus (-) binary operator to subtract two two objects by subtracting their data members and return the result in new object, remember value of DM's should remain legal, otherwise assign default value

//Write static member function to print information about class "This is a fuzzy class"

//Write post increment operator ++ to add 1 into both data members, if value is less than 2, otherwise don't add anything

c. Consider class data members and output (inside the box), write member function show? [5]

```
class ABC{
    int age; //for less than 10 print Invalid in show function
    char status; //E for Enable, D for Disable
    int count; //Count of values in x1 & x2
    int *x1, *x2;
```

Age: 12 years	
Status: Enable	Count: 3
X1	X2
-----	-----
23	18
126	29
34	108

d. Define following briefly:

[20]

- Typical getter function for a data member named **center** of type **float** in a class **Line**.
- Typical setter function for a data member of **Set** class named **size** of type **int**. Setter function must implement a check on **size** as non-negative.
- Definition of a destructor of **File** class to close the file pointed by a **data member** (ofstream objects) named **outputFile**
- Two different prototypes declarations (only not definition) of the Copy constructor for a class named **Cashmemo**

- e. Write the definition of the explicit operator **!=** for class named **RationalNumber** composed two integers **num** and **den**. An example rational number is at right whose **double** equivalent is **num/den**.
- f. Write definition for the overloading operator **<=** as a member **ComplexNumber** class. Two complex numbers have to be compared through their modulus ($\sqrt{x^2 + y^2}$).

- g. Complete following code to save object **obj** in both files.

```
int main(){
    ofstream out1("data.txt"); //text file
    ofstream out2("data.bin", ios::binary);
    Simple obj(...);
```

```
class Simple{
public:
    int x;
    float f;
    double d;
};
```

- h. If member function perfectly **copy** (making deep copy) for class **Klacc** is available, write the definition of **copy constructor** & **assignment operator** using **copy** function?

```
class Klacc{
    //Use for dynamic arrays
    int *c; // for 1d
    float **d; // for 2d
};
```

- i. Give an example with necessary code of any constructor of class named **AAA**, to initialize its **const data member CC** of type **double** with the value provided as a parameter.